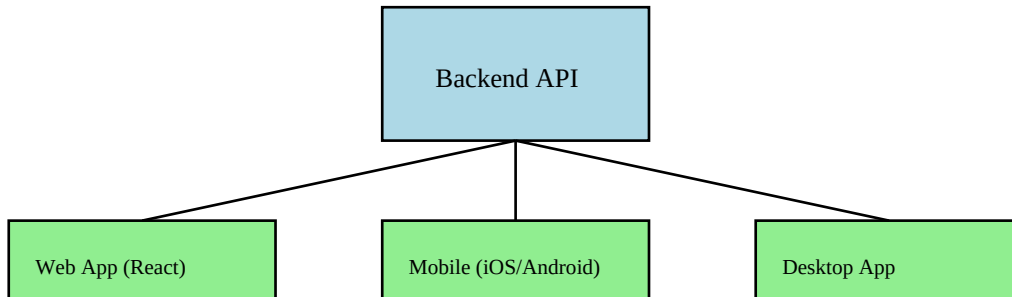


■ API-First Backend Architecture

This document explains the concept of building a **single backend** that exposes REST API routes, and having **multiple platform clients** (web, mobile, desktop) consume those APIs. This approach is widely adopted in modern MERN/PERN and SaaS applications.



- Backend exposes REST API routes like `/api/users`, `/api/login`.

- JSON is the universal format for responses: `{ id: 1, name: 'Alice' }`.

- All clients (web, mobile, desktop) can consume the same API.

- JWT Authentication is widely used for secure access.

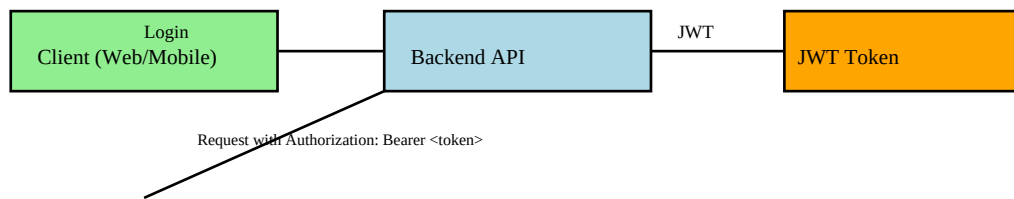
- Single backend, many frontends → scalable and future-proof.

Example Code:

```
// Example API (Express.js)
app.get("/api/users", (req, res) => {
  res.json([ { id: 1, name: "Alice" }, { id: 2, name: "Bob" } ]);
});

app.post("/api/login", (req, res) => {
  res.json({ token: "jwt-token-here" });
});
```

JWT Authentication Flow:



This **API-first** approach allows you to build your backend once and re-use it across all platforms. It reduces duplication, improves scalability, and ensures consistent behavior across devices.